

The matrix of linear map

Suppose $V(F)$ and $W(F)$ are finite dimensional vector spaces with bases $\{v_1, v_2, \dots, v_n\}$ and $\{w_1, w_2, \dots, w_m\}$, respectively.

Then let $T: V \rightarrow W$ be a linear map.

Then the matrix of T , having entries $a_{ij} \in F$ is defined by

$$T(v_j) = a_{1j}w_1 + a_{2j}w_2 + \dots + a_{mj}w_m$$

for j^{th} column

for $j = 1, 2, \dots, n$

It shows that j^{th} column of matrix A of linear map T consists of the coordinates of $T(v_j)$ in the chosen basis for W .

Every matrix $A \in \mathbb{R}^{m \times n}$ induced a linear map

$T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$T(v) = Av$$

Example

L8-1b

Consider linear map $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$T(x_1, x_2) = (2x_1 - 3x_2, 4x_1 + 5x_2)$$

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 & -3 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Find the matrix representation of T relative to

basis $B = \{(1, 2), (3, 5)\}$

$$T(1, 2) = (-4, 14) = a_{11}(1, 2) + a_{21}(3, 5)$$

$$T(3, 5) = (-9, 37) = a_{12}(1, 2) + a_{22}(3, 5)$$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$$T(x_1, x_2) = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$= (2x_1 + x_2, 4x_1 + 3x_2)$$

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T: \mathbb{R}^m \rightarrow \mathbb{R}^n \rightarrow \begin{bmatrix} \end{bmatrix}_{m \times n \text{ matrix}}$$

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

Find standard matrix for transformation T .

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 \\ 3x_3 \end{bmatrix}$$

Let say we have basis $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$T(v_1) = \begin{bmatrix} 1-0 \\ 3(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$T(v_2) = \begin{bmatrix} 0-1 \\ 3(0) \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$T(v_3) = \begin{bmatrix} 0-0 \\ 3(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Using these as our columns, the standard matrix for T is

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\begin{aligned} T(x) &= \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} x_1 - x_2 \\ 3x_3 \end{bmatrix} \end{aligned}$$

For any linear transformation T b/w $\mathbb{R}^m \rightarrow \mathbb{R}^n$, for some m, n , one can find a matrix which implements the mapping.

So;

One can say that linear map can be encoded with a matrix, and every matrix-vector multiplication can be interpreted as applying a linear map to a vector.

* Every linear map preserves lines:

Linear Maps as Geometric Transformation

Every linear map can be decomposed into the compositions of dilatations, rotations, and reflections.

[will be studied when we go through Singular Value Decomposition]

Dilatation

Any function that operates on vectors by stretching or shrinking vector is a linear map whose

this gives by

$$\begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & \lambda_2 & 0 & 0 \\ 0 & 0 & \lambda_3 & 0 \\ 0 & 0 & 0 & \lambda_n \end{bmatrix}$$

for positive λ_i scalars

L8-1e

Rotations.

Any function that rotates vectors by some angle θ about origin is a linear map.

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Reflections

Any function that reflects vectors across a line passing through the origin is ~~to change coordinates~~ ~~to a basis~~ a linear map.

Example. Matrix for a reflection across the x -axis

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

L8-2a

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3 \times 4} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}_{4 \times 1} = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix}_{3 \times 1}$$

$$\mathbb{R}^4 \rightarrow \mathbb{R}^3$$

One can simplify & say

Using linear map one can modify dimensions and project into other dimensions.

Nullspace

Let $T: V \rightarrow W$ be a linear map. Then the nullspace denoted by $\text{null}(T)$, is the subset of V consisting of those vectors that T maps to 0 .

$$\text{null}(T) = \{ v \in V \mid T(v) = 0 \}$$

v belongs to V s.t. $T(v) = 0$.

Some people use the term kernel instead of nullspace.

Null is zero (remember)

* The nullspace is a subspace.

Suppose $T: V \rightarrow W$. Then $\text{null}(T)$ is a subspace of V .

Range

L8-26

Let $T: V \rightarrow W$ be a linear map.

Then the range of T is the subset of W consisting of those vectors that are the form Tv for some $v \in V$.

$$\text{range}(T) = \{w \in W \mid \exists v \in V \text{ s.t. } T(v) = w\}$$

w belongs to W such that there exists v belongs to V such that $T(v) = w$.

$$\text{ray}(T) = \{Tv : v \in V\}$$

Example - $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

$$T(x, y) = (2x, 5y, x+y).$$

$$\text{range } T = \{(2x, 5y, x+y) \mid x, y \in \mathbb{R}\}.$$

A basis of $\text{ray } T$ is

$$(2, 0, 1), (0, 5, 1)$$

▷ Range of each linear map is a subspace of the vector space into which it is being mapped.

$T: V \rightarrow W$; then $\text{range}(T)$ is a subspace of W

~~Some people~~

Some people
call it
Image.

L8-3a

Fundamental Theorem of Linear Maps. Rank-Nullity Theorem.

Let $T: V \rightarrow W$ be a linear map.

Then $\text{range}(T)$ is finite-dimensional

and

$$\dim V = \dim \text{range}(T) + \dim \text{null}(T).$$

Ex

Determine range & null space of

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^4$$

$$T(x_1, x_2, x_3) = (x_1 - x_2 + x_3, x_2 - x_3, x_1, 2x_1 - 5x_2 + 5x_3)$$

$$\text{Range } T: T(1, 0, 0) = (1, 0, 1, 2) \rightarrow a$$

$$T(0, 1, 0) = (-1, 1, 0, -5) \rightarrow b$$

$$T(0, 0, 1) = (1, -1, 0, 5) \rightarrow c$$

b & c are linearly dependent as they are -1.

$$\text{Range}(T) = \text{span}\{(1, 0, 1, 2), (-1, 1, 0, -5)\}$$

$$\text{Rank}(T) = 2$$

$$\text{Nullspace}(T): T(x) = 0$$

$$x_1 - x_2 + x_3 = 0$$

$$x_2 - x_3 = 0$$

$$x_1 = 0$$

$$2x_1 - 5x_2 + 5x_3 = 0$$

$$x_1 = 0 \text{ so}$$

$$-x_2 + x_3 = 0$$

$$x_2 - x_3 = 0$$

$$-5x_2 + 5x_3 = 0$$

$$x_2 = x_3$$

$$\{(0, 1, 1)\}$$

$$\text{nullity}(T) = 1$$

Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$

L8-3b

$$T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, 2x_1 + x_2)$$

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} =$$

Initial basis e_1, e_2 standard basis for $\mathbb{R}^2$

Final basis e'_1, e'_2, e'_3 standard basis for $\mathbb{R}^3$

$$T(e_2) = T(0, 1) = (1, -1, 1) = 1e'_1 - 1e'_2 + 1e'_3$$

$$T(e_1) = T(1, 0) = (1, 1, 2) = 1e'_1 + 1e'_2 + 2e'_3$$

So matrix $(T)_{e'_i}^e$ is
$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

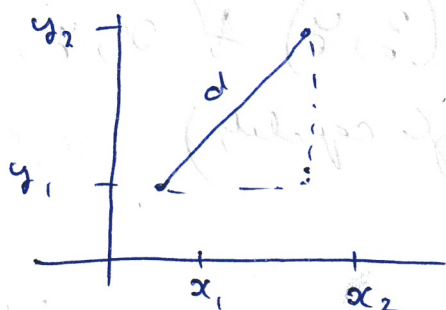
LS-4a

Norm & Space

These spaces are needed to make in depth analysis of some of ML algorithms

The distance metric helps algorithms to recognize similarities b/w the contents

Easiest one can think of is



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

A distance function provides distance b/w the elements of a set

If distance is zero then elements are equivalent else they are different from each other

Metric

L8-4b

A metric on a set S is a function

$$d: S \times S \rightarrow \mathbb{R}$$

satisfying properties

Non-negativity

$$(M1) \quad d(x, y) \geq 0 \quad \forall x, y \in S \quad \xrightarrow{\text{Non-negativity}} \quad d(x, y) = 0 \\ \text{iff } x = y \quad [\text{Identity}]$$

$$(M3) \quad d(x, y) = d(y, x) \quad (\text{Symmetry})$$

$$(M4) \quad d(x, y) \leq d(x, z) + d(z, y) \quad \forall x, y, z \in S \\ (\text{Triangle equality})$$

Distance that satisfies (M1) (M3) and (M4) ^{but} _{not} (M2) is called pseudometric

Distance that satisfies (M1), (M2) and (M4) (but not necessarily (M3)) is called quasimetric

~~Stop~~

Norm

A norm on a real vector space V is a function

$$\|\cdot\|: V \rightarrow \mathbb{R}$$

such that

- ① $\|x\| \geq 0 \quad \forall x \in V$ and $\|x\| = 0 \iff x = \vec{0}$
- ② $\|\alpha x\| = |\alpha| \|x\| \quad \forall x \in V$ and $\alpha \in \mathbb{R}$
- ③ $\|x+y\| \leq \|x\| + \|y\| \quad \forall x, y \in V$

$(V, \|\cdot\|)$ is called normed space or norm vector space or norm linear space

* Every normed space is metric space.

→ Converse is not true.

Examp^l. of some norms on $\mathbb{R}^n$

→ $\|x\|_1 = \sum_{i=1}^n |x_i|$ L_1 norm → Manhattan norm

→ $\|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$ L_2 norm - Euclidean norm

→ $\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$ L_p norm or p norm

Dot product

→ $\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$ ∞ norm

1. Manhattan norm & L_2 distance

Ex let $v \in \mathbb{R}^3$

$$x = (1, 0, -3)$$

$$\|x\|_1 = 1 + 0 + |-3| = 4$$

Manhattan norm

$$\|x\|_2 = (1^2 + 0^2 + (-3)^2)^{1/2} = \sqrt{10}$$

L_2 norm
Euclidean

$$\|x\|_\infty = 3$$

Distance b/w 2 vectors

Consider

$$x = (x_1, x_2, x_3, \dots, x_d)$$

$$y = (y_1, y_2, y_3, \dots, y_d) \in \mathbb{R}^d$$

L_p distance

$$d_p(x, y) = \|x - y\|_p = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$$

Ex Let $v \in \mathbb{R}^3$

$$X = (1, 0, -3)$$

$$\|X\|_1 = 1 + 0 + |-3| = 4$$

Manhattan norm

$$\|X\|_2 = (1^2 + 0^2 + (-3)^2)$$

$$= \sqrt{10}$$

L_2 norm, Euclidean

$$\|X\|_\infty = 3$$

gf distance b/w 2 vectors

Consider

$$x = (x_1, x_2, x_3, \dots, x_d)$$

$$y = (y_1, y_2, y_3, \dots, y_d)$$

in $\mathbb{R}^d$

$$d_p(x, y) = \|x - y\|_p =$$

$$\left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$$

L8-6a

Most common is L_2

$$d_2(x, y) = \|x - y\| = \|x - y\|_2 = \sqrt{\sum_{i=1}^d (x_i - y_i)^2}$$

Euclidean norm $\rightarrow$ straight line

* L_1 ("Manhattan" distance)

$$d_1(x, y) = \|x - y\|_1 = \sum_{i=1}^d |x_i - y_i|$$

Reason Manhattan, as you will stay on street, can't cut through buildings

Named

* L_0 distance

$$d_0(x, y) = \|x - y\|_0 = d - \sum_{i=1}^d \mathbb{1}(x_i = y_i)$$

$$\text{where } \mathbb{1}(x = y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$$

d_0 is not convex

L_∞ distance

$$d_\infty(x, y) = \|x - y\|_\infty = \max_{i=1, \dots, d} |x_i - y_i|$$

Max deviation along any one coordinate.

d_∞ is not technically L_p distance, but is limit of such distance

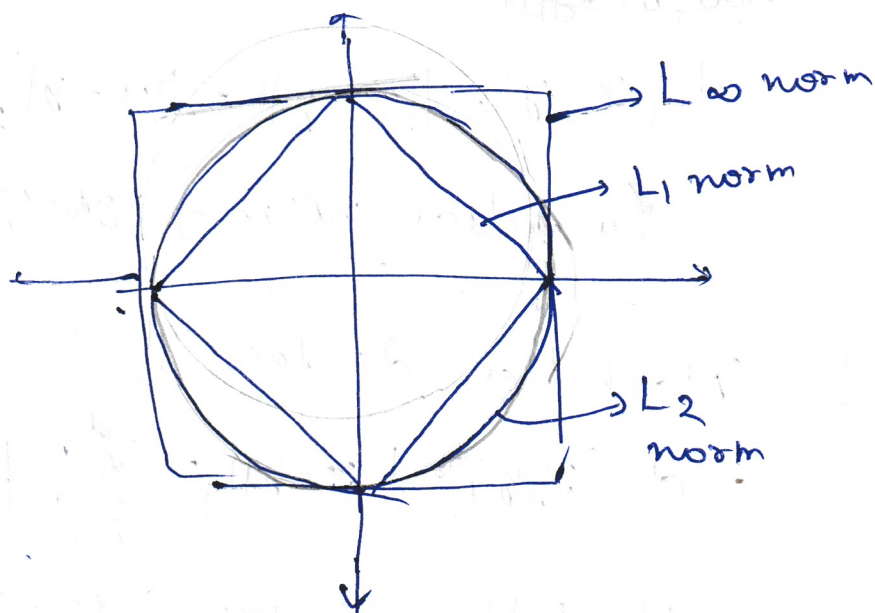
d_∞ is not technically L_p distance, but is the limit of such distance as p goes to ∞ .

Geometry

L_1, L_2 & L_∞
in $\mathbb{R}^2$.

L_1 is never larger than L_2 .

L_2 is never larger than L_∞ .



L_p is never greater than $L_{p'}$ for $p < p'$

→ It is possible to draw L_p for $p < 1$

However these are not convex and they curve in b/w the coordinate axes

Convex function

A function $f: S \subseteq \mathbb{R}^n$ is said to be convex if for $x_1, x_2 \in S$; we have

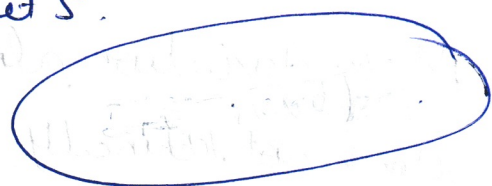
$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda)f(x_2)$$

where $0 \leq \lambda \leq 1$

Convex set

A set is said to be convex if the line joining any two points of set lies entirely on set S .

$x_1, x_2 \in S$



L zero

L_0 norm $\rightarrow \|X\|_0$ is the no. of non zero elements of X

$$X = (5, 6, 0, 0, 0, 10, 7, 0) \in \mathbb{R}^8$$

$$\text{then } \|X\|_0 = 4$$

$\rightarrow$ Important in compressing sparse soln.

It is not a norm due to second

$$\|\alpha X\| \neq \|\alpha\| \|X\| \quad \text{for } \alpha \neq 1$$

$$\alpha = 2$$

L.H.S.)

R.H.S.)

$\downarrow$
Star
4

$\neq$

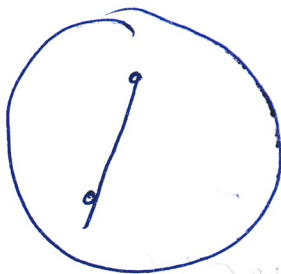
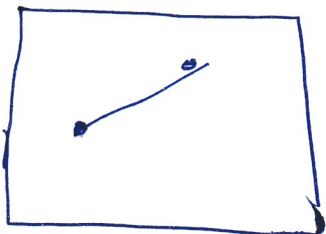
$$2 \cdot 4 = 8$$

L8-7a

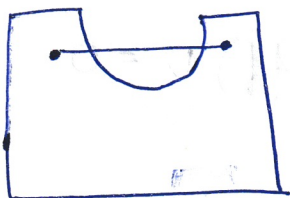
$x_1, x_2 \in S$

Convex Set

$$\Rightarrow \lambda x_1 + (1-\lambda)x_2 \in S$$



Line joining lies
inside ~~set~~ the set
convex set



not inside the set
so not convex

Inner Product space

L8-7b

An inner product on a real vector space
is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$$

satisfying

i) Positivity

$$\langle v, v \rangle \geq 0 \quad \forall v \in V$$

ii) Definiteness

$$\langle v, v \rangle = 0 \text{ if and only if (iff) } v = 0$$

iii) Additivity in first slot

$$\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle \text{ for all } u, v, w \in V.$$

iv) homogeneity in first slot

$$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle \text{ for all } \lambda \in \mathbb{R}.$$

v) Conjugate Symmetry

$$\langle u, v \rangle = \overline{\langle v, u \rangle} \text{ for all } u, v \in V.$$

for complex

$$\langle u, v \rangle = \overline{\langle v, u \rangle}$$

L8-8a

$$V = \mathbb{R}^n(\mathbb{R})$$

$$\langle x, y \rangle = \sum_i x_i y_i$$

$$= x_1 y_1 + x_2 y_2 + \dots + x_n y_n$$

$x \cdot y \rightarrow$ usual dot product

inner product space

$$\langle x, y \rangle = \|x\| \|y\| \cos \theta$$

Orthogonal Vectors

In an inner product vector space
two vectors u, v are said to be orthogonal

$$\Leftrightarrow \langle u, v \rangle = 0$$

Simplest examples

$(1, 0)$ & $(0, 1)$ are orthogonal in $\mathbb{R}^2(\mathbb{R})$

Most famous theorem based on a Pythagorean theorem

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2$$

$$\|u + v\|^2 = \langle u + v, u + v \rangle$$

$$\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$= \|u\|^2 + \|v\|^2$$

Orthogonal Complement

L8-8b

If $W \subseteq V$; where V is the inner product space, then the orthogonal complement of W ; denoted by $W^\perp$, is the set of vectors in V that are orthogonal to every element of W .

$$W^\perp = \{ v \in V \mid v \perp w \forall w \in W \}$$

Simplest

$$V = \mathbb{R}^3 (\mathbb{R})$$

$$W = \{ (1, 0, 0), (0, 1, 0) \}$$

$$W^\perp = \{ (0, 0, 1) \}$$

Orthogonal Projection

The linear transformation

$$P: V \rightarrow W$$

is called the orthogonal projection onto

W .

Orthogonal Projection

Let V be an inner product space and W be a sub-space of V .

Also let $\{w_1, w_2, \dots, w_k\}$ be an orthogonal basis of W .

Then linear transformation

$$v \in V$$

$P: V \rightarrow W$ defined as

$$P_W(v) = \frac{\langle v, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 + \frac{\langle v, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 + \dots + \frac{\langle v, w_k \rangle}{\langle w_k, w_k \rangle} w_k$$

is called orthogonal projection of W .

As any linear map P that satisfy $P^2 = P$ is called a projection.

so P_W is a projection!

Let $w_1 = (1, 0, 1)$ and

L8-9b

$$w_2 = (0, 3, 0).$$

define $W = L(w_1, w_2)$

Take a vector $v = (0, 5, 10) \in \mathbb{R}^3$

$$P_W(v) = \frac{\langle v, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 + \frac{\langle v, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

$$= \frac{10}{2} (1, 0, 1) + \frac{15}{9} (0, 3, 0)$$

$$= (5, 0, 5) + (0, 5, 0)$$

$$P_W(v) = (5, 5, 5)$$

Project of v on subspace W

$$v - P_W(v)$$

$$= (0, 5, 10) - (5, 5, 5) = (-5, 0, 5) \in W^\perp$$
